MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSEECT203 AIandMachineLearning Fundamentals

Duration:2hours **MaximumMarks:70**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I)and Appreciation (Ap)*

Part A

MultipleChoiceQuestions,Answer**All**Questions

Eachquestioncarries1mark

1	WhatistheprimarygoalofArtificialIntelligence(AI)? A) Tobuildwebsitesusingcode B) Toanalyzebusinessfinances C) Toenablemachinestomimichumanintelligence D) Toimproveinternetspeed	U	1
2	Whichofthefollowingbestdescribesanintelligentagent? A) Acomputerthatrunswithout electricity B) Amachinethatperformsrepetitivetasksonly C) Anentitythatperceivesitsenvironmentandtakesactionstomaximizesuccess D) Apersonwhowritesmachine code	U	1
3	HowdoesMachineLearning(ML)differfromtraditionalprogramming? A) MLdoesnotrequire data B) InML,systemslearnfromdatainsteadofbeingexplicitlyprogrammed C) MLusesonlyExcel formulas D) MLavoidsusing algorithms	U	1
4	WhichofthefollowingisanexampleofMLinhealthcare? A)Videoconferencing B)Diseasepredictionusingpatientdata C)Onlinebill payment D)Voicecallingthroughmobilenetworks	U	1
5	WhatisthemaindifferencebetweenMLandDeepLearning(DL)? A) DLusesmanualrulesonly B) DLisnotapartofML C) DLusesneuralnetworkswithmultiplelayersforlearning D) DLrequiresnodataatall	U	1
6	WhichPythonlibraryismainlyusedfornumericalcomputations? A)Matplotlib B)Pandas C)NumPy D)Seaborn	A	2
7	WhatistheprimarypurposeofthePandaslibraryinPython? A)Performingimageprocessing B)Creatinganimations C)Datamanipulationandanalysis D)Designingwebsites	A	2
8	Which tool provides a web-based interface for writing and running Python code, commonly used in data science? A) AndroidStudio B)JupyterNotebook C)Notepad++ D)Excel	A	2
9	WhatisthepurposeofExploratoryDataAnalysis(EDA)? A) Todesignmachinelearningalgorithms	A	2

	B) Tocleanthedataonly C) Tosummarizeandvisualizedatatodiscoverpatterns D) TocompilePythoncode		
10	InPython,whichcommandisusedtoreadaCSVfileusingPandas? A)numpy.read_csv() B)pandas.load() C) read.csv() D)pd.read_csv()	A	2
11	Whichofthefollowingisatypeofsupervisedlearningalgorithm? A)K-MeansClustering B)HierarchicalClustering C)Linear Regression D)AprioriAlgorithm	A	3
12	WhatisthemainobjectiveofK-Meansclustering? A)Predictfuturevalues B)Classifylabeleddata C)Groupsimilardatapointsintoclusters D)Minimizeclassificationerror	A	3
13	InK-NearestNeighbors(KNN),thevalueof'K'refers to: A) Thenumberofinputfeatures B) Thenumberofoutput classes C) Thenumberofnearestneighborsconsideredforclassification D) Thenumberofhiddenlayers	A	3
14	Whichmachinelearningtechniqueisbasedonrewardandpunishmentmechanism? A)SupervisedLearning B)UnsupervisedLearning C)ReinforcementLearning D)DeepLearning	A	3
15	Whatisthegoal ofhierarchical clustering? A)Reducedimensionalityofdata B)Predictnumericalvalues C)Groupdataintoatreeofclusters D)Improveimagequality	A	3
16	Whatisthemainfunctionofanactivationfunctionlikethesigmoidinaneural network? A) Tostoreinput data B) Toconvertdatatobinary C) Tointroducenon-linearityintothe model D) Toincreasethenumberofneurons	U	4
17	Whatarethetwokeyparametersusedinaneuralnetwork model? A)EpochsandSteps B)LayersandNodes C)WeightsandBiases D)LearningRateandMomentum	U	4
18	Inthegradientdescentalgorithm,theroleofthelearningrateisto: A)Savethetrainingmodel B)Measuremodelaccuracy C)Controlthestepsizeinweightupdates D)Countthenumberoflayers	U	4
19	Theperceptronisbestdescribedas: A) Aclustering algorithm B)Asingle-layerbinaryclassifier C)Adatavisualizationtool D)Abackpropagationmethod	U	4
20	Whatisthepurposeofback-propagationintraininganeural network? A) Toinitializeweights B)Toperform classification C)Toadjustweightsusingerrorvalues D)Tovisualizeresults	U	4
			1 * 20 = 20

Part B

Short Answer Questions

Answer any **6** QuestionsEachquestioncarries5marks

21	Definean intelligentagent withan example.	U	1
22	Writeanytworeal-worldapplicationsofmachine learning.	U	1
23	WhatisthedifferencebetweenNumPyandPandasinPython?	A	2

24	Name any two tools used to set up Machine Learning environment and explain briefly.	A	2
25	Definesupervisedlearning andgiveone example.	A	3
26	Whatisreinforcementlearning?	A	3
27	Howdoesabiologicalneurondiffer froman artificial neuron?	U	4
28	Whatisacostfunction inaneuralnetwork, andwhy isit important?	U	4

6 * 5 = 30

Part C

Essay QuestionsAnswerany**2**Questions Eachquestioncarries10marks

29	Explainthe differencesbetweenArtificialIntelligence,Machine Learning, and Deep Learning. Illustrate with suitable examples.	U	1
30	Explain the steps involved in data pre-processing and feature engineering.Why are these important in machine learning?	A	2
31	Compare and explain the three main types of machine learning: Supervised, Unsupervised, and Reinforcement Learning. Give suitable examples.	A	3
32	Explainthestructureandworkingofafeed-forwardneuralnetwork.How do forward propagation and gradient descent help in model training?	U	4

2 * 10 = 20